Poisson Regression

Supplement

Rebecca C. Steorts (slide adaption from Maria Tacket) and material from Chapter 4 of Roback and Legler text.

Learning goals

- Recognize when **Poisson regression** is appropriate for count data
- Understand the **mean = variance** property and why it can fail
- Explain the motivation for **excess zeros**
- Describe the structure of a **Zero-Inflated Poisson (ZIP)** model
- Interpret coefficients in the **count** and **zero** components

Count data: when do we use Poisson models?

We model a non-negative integer response:

$$Y \in \{0, 1, 2, \ldots\}$$

Examples

- ER visits in a year
- Defects per item
- Emails per day
- Crashes at an intersection per month

The Poisson distribution (review)

$$P(Y = y) = \frac{e^{-\lambda}\lambda^y}{y!}, \qquad y = 0, 1, 2, \ldots$$

Key properties:

$$\mathbb{E}(Y) = \lambda, \qquad \mathrm{Var}(Y) = \lambda$$

So **mean = variance**.

Poisson regression (GLM)

Model the mean count λ_i as a function of predictors:

$$Y_i \sim \text{Poisson}(\lambda_i), \qquad \log(\lambda_i) = \mathbf{x}_i^\top \boldsymbol{\beta}$$

Why the log link?

- ensures $\lambda_i > 0$
- leads to a clean multiplicative interpretation

Interpreting coefficients in Poisson regression

From

$$\log(\lambda_i) = \beta_0 + \beta_1 x_{i1} + \cdots$$

A one-unit increase in x_j changes the mean by:

$$\frac{\lambda_i(x_j + 1)}{\lambda_i(x_j)} = e^{\beta_j}$$

So e^{β_j} is the **multiplicative factor** on the mean count.

A common issue: too many zeros

In practice, we often see **more zeros** than a Poisson model expects.

Examples

- doctor visits (many people never go)
- insurance claims (many policyholders have none)
- crashes (many locations have zero)

Consequences:

- poor fit
- underestimated standard errors
- apparent overdispersion ($\mathrm{Var}(Y) > \mathbb{E}(Y)$)

Two kinds of zeros (intuition)

Sampling zeros:

- could have been positive, but happened to be 0 this time
- e.g., someone who *might* visit a doctor this year, but didn't

Structural zeros:

- outcome is essentially impossible / not at risk
- e.g., someone without a car $\rightarrow$ 0 car crashes

ZIP explicitly models this mixture.

Zero-Inflated Poisson (ZIP): the story

For each observation i:

1. With probability π_i, we are in a **structural zero** state
2. With probability $1 - \pi_i$, we are in the **count** state:

$$Y_i \mid \text{count state} \sim \text{Poisson}(\lambda_i)$$

ZIP probability model

$$P(Y_i = 0) = \pi_i + (1 - \pi_i)e^{-\lambda_i}$$

$$P(Y_i = y) = (1 - \pi_i)\frac{e^{-\lambda_i}\lambda_i^y}{y!}, \quad y \geq 1$$

So ZIP adds a **spike at zero** on top of the Poisson distribution.

ZIP has two linked regressions

Zero (inflation) part: models probability of structural zero

$$\text{logit}(\pi_i) = \mathbf{z}_i^\top \boldsymbol{\gamma}$$

Count part: models mean count among the "at risk" group

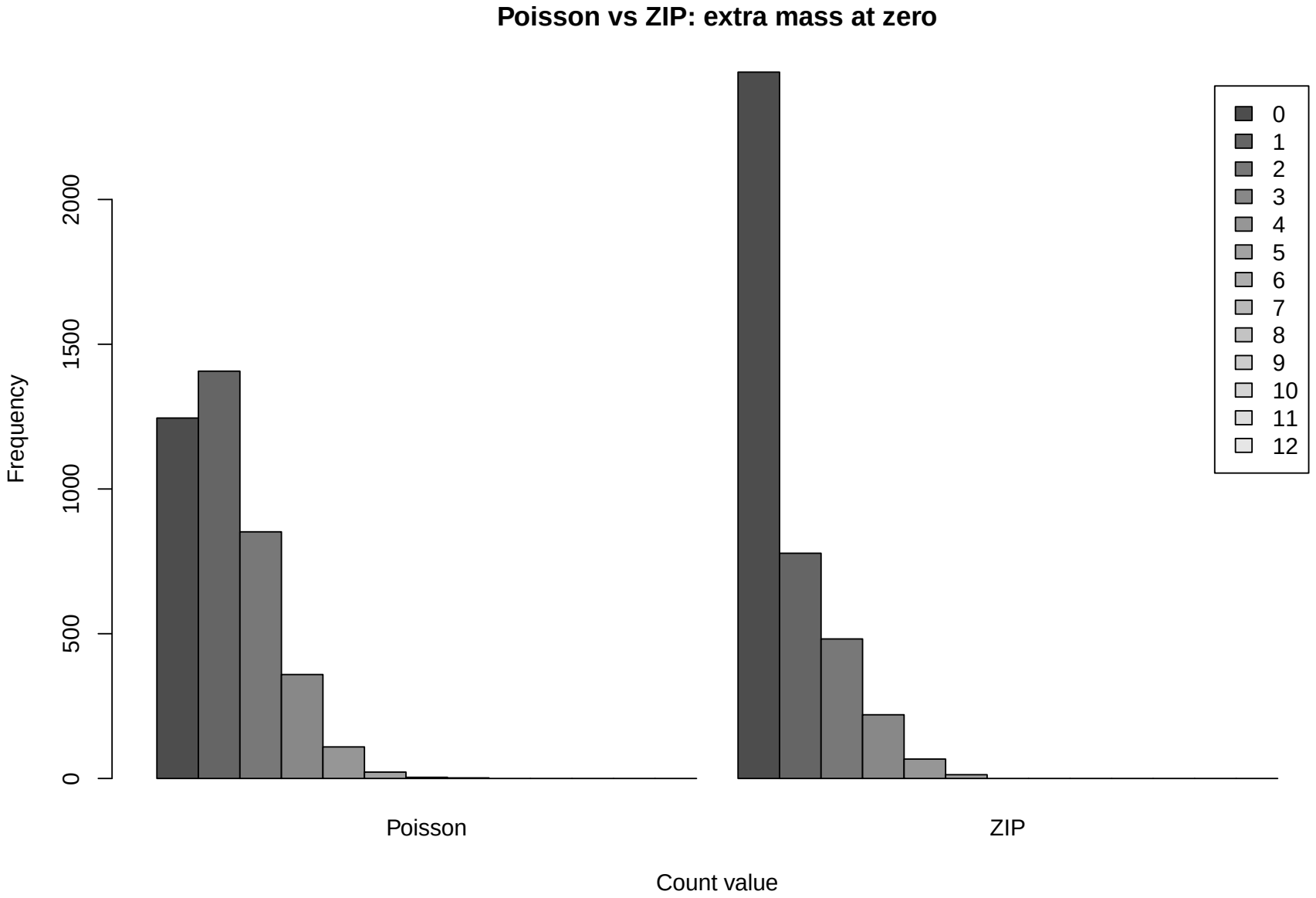
$$\log(\lambda_i) = \mathbf{x}_i^\top \boldsymbol{\beta}$$

Predictors $\mathbf{x}_i$ and $\mathbf{z}_i$ can be the same or different.

Interpreting ZIP coefficients

- **Count coefficients** β:
  Among those in the count state, a one-unit increase in x_j multiplies the mean by e^{β_j}.
- **Zero coefficients** γ:
  In the logit model, a one-unit increase in z_k multiplies the **odds** of being a structural zero by e^{γ_k}.

Visual intuition: Poisson vs ZIP



What ZIP is *not*

ZIP is not just "overdispersion" in general.

If variance $>$ mean but zeros are not *structural*, consider:

- **Negative Binomial regression** (handles heterogeneity / overdispersion)
- **Quasi-Poisson** (robust SEs, not a full likelihood model)

ZIP is best when **two mechanisms** generate zeros.

How do we decide between Poisson and ZIP?

Practical checks:

- Compare observed zeros vs expected zeros under Poisson
- Residual diagnostics / goodness of fit
- Likelihood-based comparisons (AIC/BIC) for fitted models
- Subject-matter reasoning: "not-at-risk" group plausible?

Example (conceptual)

Suppose Y = number of clinic visits in a year.

- Some people are "never-users" (structural zeros)
- Others are at risk and have counts driven by health and access

ZIP lets you model:

- who is a never-user (zero model)
- how often users visit (count model)

Fitting ZIP models in R

The `pscl` package provides `zeroinfl()`.

```
# install.packages("pscl")
library(pscl)

# Example:
# y ~ x1 + x2 is the count model
# | z1 + z2 is the zero-inflation model
fit_zip <- zeroinfl(y ~ x1 + x2 | z1 + z2,
                    dist = "poisson")
summary(fit_zip)
```

Key takeaways

- **Poisson regression** models counts with $\mathbb{E}(Y) = \mathrm{Var}(Y)$
- Real data often have **excess zeros**
- **ZIP** combines:
  - a logistic model for structural zeros (π_i)
  - a Poisson model for counts among at-risk units (λ_i)
- Interpret coefficients separately in the two parts